Graphs of Inverse Trigonometric Functions & Their Derivatives (AA HL)

ClipSAT Printable Practice Suite (clipsat.org) • Topic 5.04

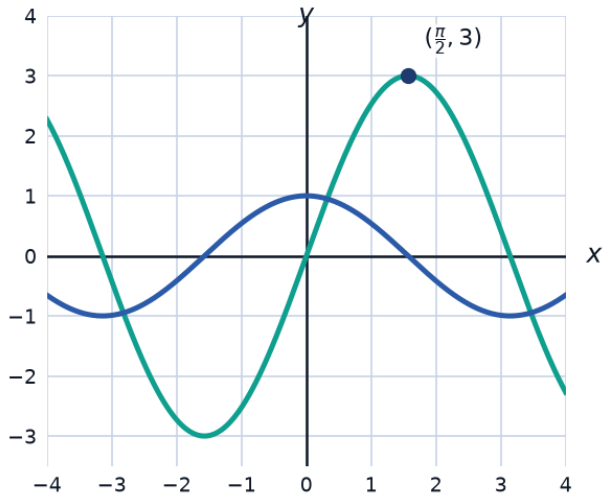
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Domains and Ranges of Inverse Trig Functions**

Consider the complex number z such that $z = a + 5i$ and $|z| = \sqrt{26}$. Given that a is a real number, find the possible values of a.

2 **[Foundational]** • **Target: Derivatives of Inverse Trigonometric Functions**

A factory uses machines M_1, M_2, M_3 producing $30\%, 30\%, 40\%$ of items respectively, with defect rates $\frac{2}{5}, \frac{2}{5}, \frac{3}{10}$. Given an item is defective, find $P(M_2|\text{defective})$, to 3 decimal places.



3 **[Foundational]** • **Target: Tangent Lines and Extreme Values**

A factory uses machines M_1, M_2, M_3 producing $30\%, 30\%, 40\%$ of items respectively, with defect rates $\frac{3}{10}, \frac{1}{10}, \frac{2}{5}$. Given an item is defective, find $P(M_1|\text{defective})$, to 3 decimal places.

4 **[Foundational]** • **Target: Integration Producing Inverse Trig Functions**

A factory uses machines M_1, M_2, M_3 producing $30\%, 30\%, 40\%$ of items respectively, with defect rates $\frac{3}{10}, \frac{2}{5}, \frac{1}{10}$. Given an item is defective, find $P(M_2|\text{defective})$, to 3 decimal places.

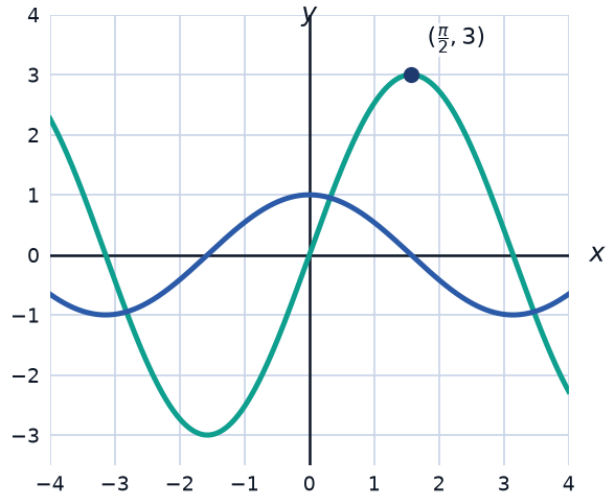
5 **[Foundational]** • **Target: Domains and Ranges of Inverse Trig Functions**

A factory uses machines M_1, M_2, M_3 producing $30\%, 30\%, 40\%$ of items respectively, with defect rates $\frac{2}{5}, \frac{2}{5}, \frac{3}{10}$. Given an item is defective, find $P(M_1|\text{defective})$, to 3 decimal places.

© ClipSAT — clipsat.org | All Rights Reserved

6 [Foundational] • Target: Derivatives of Inverse Trigonometric Functions

A factory uses machines M_1, M_2, M_3 producing 30%, 30%, 40% of items respectively, with defect rates $\frac{1}{5}, \frac{2}{5}, \frac{3}{10}$. Given an item is defective, find $P(M_3|\text{defective})$, to 3 decimal places.



7 [Foundational] • Target: Tangent Lines and Extreme Values

A factory uses machines M_1, M_2, M_3 producing 30%, 30%, 40% of items respectively, with defect rates $\frac{1}{5}, \frac{2}{5}, \frac{3}{10}$. Given an item is defective, find $P(M_2|\text{defective})$, to 3 decimal places.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Integration Producing Inverse Trig Functions

A factory uses machines M_1, M_2, M_3 producing 30%, 30%, 40% of items respectively, with defect rates $\frac{3}{10}, \frac{1}{5}, \frac{2}{5}$. Given an item is defective, find $P(M_2|\text{defective})$, to 3 decimal places.

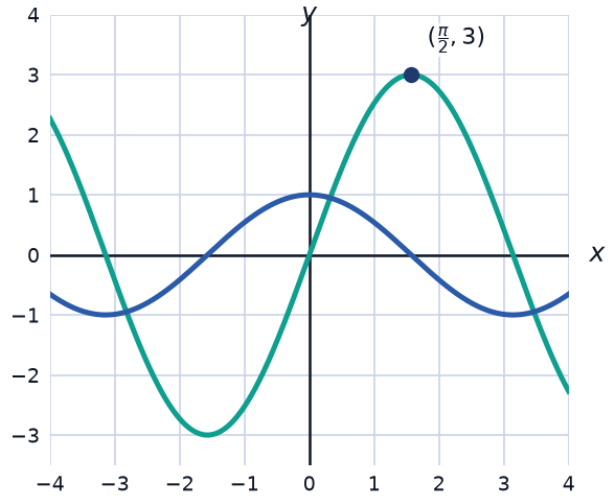
9 [Intermediate] • Target: Domains and Ranges of Inverse Trig Functions

A factory uses machines M_1, M_2, M_3 producing 30%, 30%, 40% of items respectively, with defect rates $\frac{1}{10}, \frac{1}{5}, \frac{2}{5}$. Given an item is defective, find $P(M_2|\text{defective})$, to 3 decimal places.

© ClipSAT — clipsat.org | All Rights Reserved

10 [Intermediate] • Target: Derivatives of Inverse Trigonometric Functions

A factory uses machines M_1, M_2, M_3 producing $30\%, 30\%, 40\%$ of items respectively, with defect rates $\frac{1}{10}, \frac{3}{10}, \frac{1}{10}$. Given an item is defective, find $P(M_1|\text{defective})$, to 3 decimal places.



11 [Intermediate] • Target: Tangent Lines and Extreme Values

A factory uses machines M_1, M_2, M_3 producing $30\%, 30\%, 40\%$ of items respectively, with defect rates $\frac{2}{5}, \frac{1}{10}, \frac{3}{10}$. Given an item is defective, find $P(M_2|\text{defective})$, to 3 decimal places.

12 [Intermediate] • Target: Integration Producing Inverse Trig Functions

A factory uses machines M_1, M_2, M_3 producing $30\%, 30\%, 40\%$ of items respectively, with defect rates $\frac{3}{10}, \frac{3}{10}, \frac{2}{5}$. Given an item is defective, find $P(M_2|\text{defective})$, to 3 decimal places.

13 [Intermediate] • Target: Domains and Ranges of Inverse Trig Functions

Line 1: $\mathbf{r} = (-2;\ -1;\ -5) + s(-1; 4;\ -1)$. Line 2: $\mathbf{r} = (3;\ -11;\ -12) + t(-2; 3; 4)$. Find their point of intersection.

14 [Intermediate] • Target: Derivatives of Inverse Trigonometric Functions

Line 1: $\mathbf{r} = (-2; 2;\ -1) + s(-1;\ -2; 2)$. Line 2: $\mathbf{r} = (-1; 4;\ -12) + t(0; 0;\ -3)$. Find their point of intersection.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

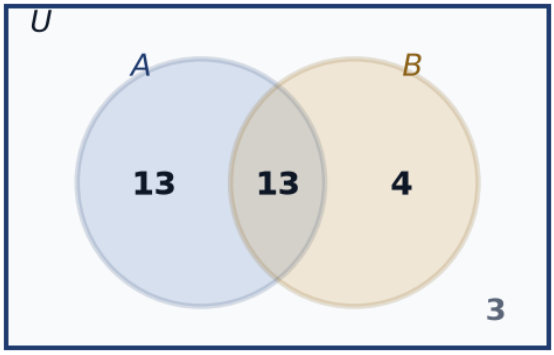
15 [Challenging] • Target: Tangent Lines and Extreme Values • Cross-Lesson: 5.01 Further Integration Techniques

Line 1: $\mathbf{r} = (-5;\ -2;\ -2) + s(-2;\ -2; 0)$. Line 2: $\mathbf{r} = (-5;\ -4; 4) + t(1; 2;\ -3)$. Find their point of intersection.

© ClipSAT — clipsat.org | All Rights Reserved

16 [Challenging] • Target: Integration Producing Inverse Trig Functions • Cross-Lesson: 5.02 Differential Equations

Line 1: $\mathbf{r} = (13;\ -13;1) + s(4;\ -3;\ -1)$. Line 2: $\mathbf{r} = (-11;8;1) + t(-4;4;\ -1)$. Find their point of intersection.



17 [Challenging] • Target: Domains and Ranges of Inverse Trig Functions • Cross-Lesson: 5.03 Maclaurin Series & Convergence of Series

Line 1: $\mathbf{r} = (-1;\ -1;16) + s(-1;0;4)$. Line 2: $\mathbf{r} = (-4;\ -7;12) + t(-3;\ -3;4)$. Find their point of intersection.

18 [Challenging] • Target: Derivatives of Inverse Trigonometric Functions • Cross-Lesson: 6.01 Matrices (AI HL)

Line 1: $\mathbf{r} = (-3;2;\ -3) + s(0;0;\ -3)$. Line 2: $\mathbf{r} = (0;\ -1;15) + t(-1;1;\ -4)$. Find their point of intersection.

19 [Challenging] • Target: Tangent Lines and Extreme Values • Cross-Lesson: 6.02 Graph Theory & Networks (AI HL)

Line 1: $\mathbf{r} = (7;7;1) + s(2;1;\ -1)$. Line 2: $\mathbf{r} = (0;8;7) + t(-1;4;3)$. Find their point of intersection.

20 [Challenging] • Target: Integration Producing Inverse Trig Functions • Cross-Lesson: 6.03 Markov Chains (AI HL)

Line 1: $\mathbf{r} = (5;\ -2;3) + s(-1;\ -3;\ -1)$. Line 2: $\mathbf{r} = (5;\ -11;15) + t(0;\ -3;4)$. Find their point of intersection.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

© ClipSAT — clipsat.org | All Rights Reserved

21 [Foundational] • Target: Domains and Ranges of Inverse Trig Functions

What are the domain and range of the principal arcsine function $y = \arcsin x$?

a. Domain: $[-1, 1]$; Range: $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

b. Domain: $(-\infty, \infty)$; Range: $[0, \pi]$

c. Domain: $[-1, 1]$; Range: $[0, \pi]$

d. Domain: $(-\infty, \infty)$; Range: $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

22 [Foundational] • Target: Domains and Ranges of Inverse Trig Functions

What are the domain and range of $y = \arccos x$?

a. Domain: $[-1, 1]$; Range: $[0, \pi]$

b. Domain: $[-1, 1]$; Range: $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

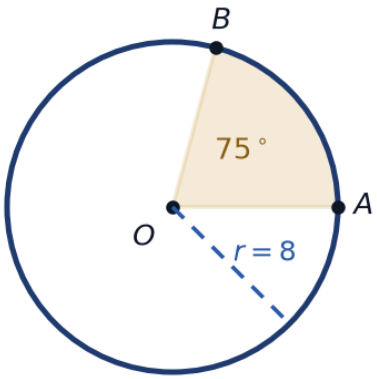
c. Domain: $(-\infty, \infty)$; Range: $[0, \pi]$

d. Domain: $[0, 1]$; Range: $[0, \pi/2]$

© ClipSAT — clipsat.org | All Rights Reserved

23 [Foundational] • Target: Domains and Ranges of Inverse Trig Functions

What are the horizontal asymptotes of the graph of $y = \arctan x$?



a. $y = \frac{\pi}{2}$ and $y = -\frac{\pi}{2}$

b. $y = \pi$ and $y = 0$

c. $x = \frac{\pi}{2}$ and $x = -\frac{\pi}{2}$

d. $y = 1$ and $y = -1$

24 [Foundational] • Target: Derivatives of Inverse Trigonometric Functions

What is the derivative of $f(x) = \arcsin(3x)$ for $|x| < \frac{1}{3}$?

a. $\frac{3}{\sqrt{1-9x^2}}$

b. $\frac{1}{\sqrt{1-9x^2}}$

c. $\frac{3}{\sqrt{1-3x^2}}$

d. $\frac{3}{1+9x^2}$

25 [Foundational] • Target: Derivatives of Inverse Trigonometric Functions

Find the derivative of $g(x) = \arctan\left(\frac{x}{2}\right)$.

a. $\frac{2}{4+x^2}$

b. $\frac{1}{1+x^2/4}$

c. $\frac{1}{4+x^2}$

d. $\frac{2}{\sqrt{4-x^2}}$

© ClipSAT — clipsat.org | All Rights Reserved

26 [Foundational] • Target: Derivatives of Inverse Trigonometric Functions

What is the relationship between the derivatives of $\arcsin x$ and $\arccos x$?

a. $\frac{d}{dx}[\arccos x] = -\frac{d}{dx}[\arcsin x] = -\frac{1}{\sqrt{1-x^2}}$

b. $\frac{d}{dx}[\arccos x] = \frac{d}{dx}[\arcsin x]$

c. $\frac{d}{dx}[\arccos x] = \frac{1}{1-x^2}$

d. They are reciprocals of each other

27 [Foundational] • Target: Tangent Lines and Extreme Values

Find the slope of the tangent line to $y = \arcsin x$ at $x = 0$.

a. 1

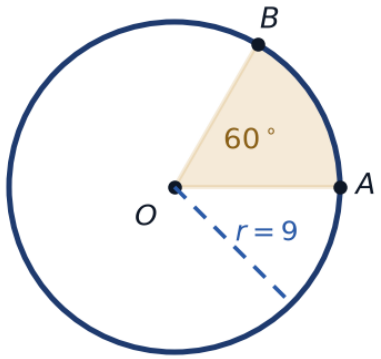
b. 0

c. $\pi/2$

d. Undefined

28 [Foundational] • Target: Tangent Lines and Extreme Values

What happens to the slope of $y = \arcsin x$ as $x \to 1^-$?



a. The slope approaches $+\infty$ (vertical tangent)

b. The slope approaches 0

c. The slope approaches 1

d. The slope approaches -1

© ClipSAT — clipsat.org | All Rights Reserved

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 **[Intermediate]** • **Target: Tangent Lines and Extreme Values**

Find the equation of the tangent line to $y = \arctan x$ at $x = 1$.

a. $y - \frac{\pi}{4} = \frac{1}{2}(x - 1)$

b. $y - \frac{\pi}{4} = 1(x - 1)$

c. $y = \frac{1}{2}x + \frac{\pi}{4}$

d. $y - 1 = \frac{1}{2}(x - \frac{\pi}{4})$

30 **[Intermediate]** • **Target: Integration Producing Inverse Trig Functions**

Evaluate $\int_0^{1/2} \frac{1}{\sqrt{1-x^2}} dx$.

a. $\frac{\pi}{6}$

b. $\frac{\pi}{3}$

c. $\frac{\pi}{4}$

d. $\frac{1}{2}$

31 **[Intermediate]** • **Target: Integration Producing Inverse Trig Functions**

Evaluate $\int_0^1 \frac{1}{1+x^2} dx$.

a. $\frac{\pi}{4}$

b. $\frac{\pi}{2}$

c. 1

d. $\ln 2$

32 **[Intermediate]** • **Target: Integration Producing Inverse Trig Functions**

Evaluate $\int \frac{1}{\sqrt{4-9x^2}} dx$.

a. $\frac{1}{3}\arcsin\left(\frac{3x}{2}\right) + C$

b. $\frac{1}{2}\arcsin\left(\frac{3x}{2}\right) + C$

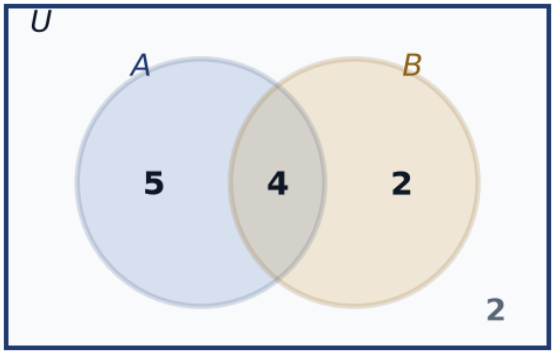
c. $\arcsin\left(\frac{3x}{2}\right) + C$

d. $\frac{1}{6}\arcsin\left(\frac{3x}{2}\right) + C$

© ClipSAT — clipsat.org | All Rights Reserved

33 [Intermediate] • Target: Domains and Ranges of Inverse Trig Functions

Line 1: $\mathbf{r} = (0;\ -1;\ -5) + s(4;\ 0;\ -2)$. Line 2: $\mathbf{r} = (2;\ 3;\ -6) + t(-2;\ -4;\ 1)$. Find their point of intersection.



a. $(1, 0, -4)$

b. $(0, -1, -5)$

c. $(-1, -3, -7)$

d. $(-2, -3, -3)$

34 [Intermediate] • Target: Derivatives of Inverse Trigonometric Functions

Line 1: $\mathbf{r} = (6;\ -5;\ 1) + s(1;\ -3;\ 1)$. Line 2: $\mathbf{r} = (2;\ 8;\ 2) + t(1;\ -4;\ -4)$. Find their point of intersection.

a. $(3, 4, -2)$

b. $(2, 3, -3)$

c. $(2, 5, -4)$

d. $(4, 2, 0)$

35 [Intermediate] • Target: Tangent Lines and Extreme Values

Line 1: $\mathbf{r} = (6;\ 4;\ -1) + s(-2;\ -1;\ 1)$. Line 2: $\mathbf{r} = (-1;\ 0;\ -2) + t(-1;\ -1;\ -4)$. Find their point of intersection.

a. $(2, 2, 1)$

b. $(-2, 0, 0)$

c. $(2, 3, 1)$

d. $(0, 1, 2)$

36 [Intermediate] • Target: Integration Producing Inverse Trig Functions

Line 1: $\mathbf{r} = (-3;\ -2;\ 5) + s(1;\ 4;\ -4)$. Line 2: $\mathbf{r} = (-2;\ 2;\ 1) + t(0;\ 0;\ 4)$. Find their point of intersection.

a. $(-3, 3, 3)$

b. $(-1, 1, -1)$

c. $(-3, 1, 2)$

d. $(-2, 2, 1)$

© ClipSAT — clipsat.org | All Rights Reserved

37 [Intermediate] • Target: Domains and Ranges of Inverse Trig Functions

Find the value of k for which $(4, 3; 4, k)$ is singular (has no inverse).

a. 1

b. -3

c. 3

d. 5

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Derivatives of Inverse Trigonometric Functions • Cross-Lesson: 6.01 Matrices (AI HL)

X takes values $0, 1, 2$ with probabilities $\frac{2}{5}, \frac{3}{10}, \frac{3}{10}$. Find $\mathrm{Var}(X)$.

a. $\frac{29}{100}$

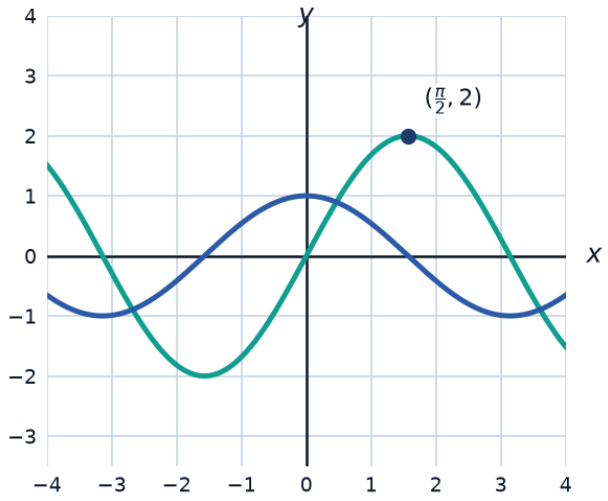
b. $\frac{101}{200}$

c. $\frac{69}{100}$

d. $\frac{59}{200}$

39 [Challenging] • Target: Tangent Lines and Extreme Values • Cross-Lesson: 6.02 Graph Theory & Networks (AI HL)

A discrete random variable X takes values $2, 3, 4$ with probabilities $\frac{3}{10}, \frac{1}{5}, \frac{1}{2}$ respectively. Find $E(X)$.



a. $\frac{16}{5}$

b. 6

c. 1

d. 4

© ClipSAT — clipsat.org | All Rights Reserved

40 [Challenging] • Target: Integration Producing Inverse Trig Functions • Cross-Lesson: 6.03 Markov Chains (AI HL)

A discrete random variable X takes values $1, 3, 5$ with probabilities $\frac{3}{10}, \frac{3}{10}, \frac{2}{5}$ respectively. Find $E(X)$.

a. $\frac{16}{5}$

b. 5

c. 0

d. 1

41 [Challenging] • Target: Domains and Ranges of Inverse Trig Functions • Cross-Lesson: 1.01 Proof by Mathematical Induction

A discrete random variable X takes values $3, 4, 5$ with probabilities $\frac{1}{5}, \frac{1}{10}, \frac{7}{10}$ respectively. Find $E(X)$.

a. 7

b. $\frac{9}{2}$

c. 1

d. 6

42 [Challenging] • Target: Derivatives of Inverse Trigonometric Functions • Cross-Lesson: 1.02 Complex Numbers: Cartesian Form & Algebra

X takes values $0, 1, 2$ with probabilities $\frac{3}{10}, \frac{1}{5}, \frac{1}{2}$. Find $\mathrm{Var}(X)$.

a. 2

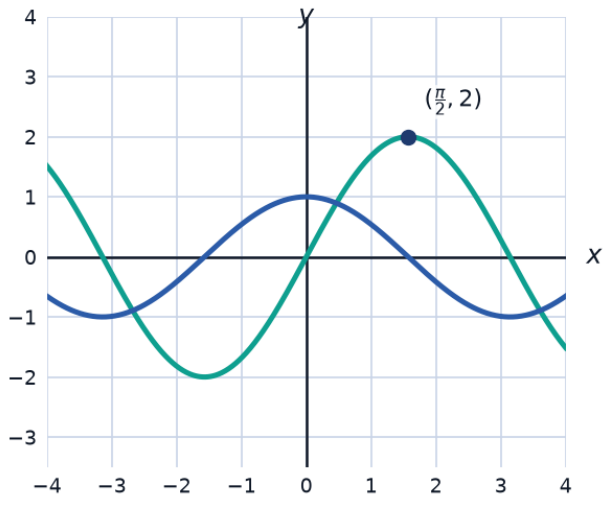
b. $\frac{39}{25}$

c. $\frac{19}{25}$

d. $\frac{49}{50}$

43 [Challenging] • Target: Tangent Lines and Extreme Values • Cross-Lesson: 1.03 Polar Form, De Moivre's Theorem & Roots of Unity

X takes values $0, 1, 2$ with probabilities $\frac{1}{5}, \frac{1}{5}, \frac{3}{5}$. Find $\mathrm{Var}(X)$.



a. $\frac{37}{50}$

b. $\frac{37}{25}$

c. $\frac{167}{100}$

d. $\frac{16}{25}$

© ClipSAT — clipsat.org | All Rights Reserved

44 [Challenging] • Target: Integration Producing Inverse Trig Functions • Cross-Lesson: 1.04 Partial Fractions

A discrete random variable X takes values $3, 4, 5$ with probabilities $\frac{1}{5}, \frac{1}{5}, \frac{3}{5}$ respectively. Find $E(X)$.

a. 7

b. $\frac{22}{5}$

c. 6

d. 2

45 [Challenging] • Target: Domains and Ranges of Inverse Trig Functions • Cross-Lesson: 1.05 Systems of Linear Equations (AA HL)

A discrete random variable X takes values $1, 2, 3$ with probabilities $\frac{1}{5}, \frac{1}{10}, \frac{7}{10}$ respectively. Find $E(X)$.

a. $\frac{5}{2}$

b. 0

c. 3

d. 5

© ClipSAT — clipsat.org | All Rights Reserved